

where S_t is some asset or combination of assets, $R(S_t)$ is the cost of the replicating portfolio and where TransactionCost represents the net costs involved in buying and selling the assets. As you are not committing any capital this is also known as riskless arbitrage—however, given the fleeting nature of such opportunities it is diminished by competition amongst arbitrageurs. Statistical arbitrage can be seen as a generalization of deterministic arbitrage, relying more heavily on the statistical properties of the mispricing dynamics. Mispricing dynamics refers to the tendency for basis risk to mean-revert or fluctuate around a stable, long-run level (Dunis et al., 2004). As such, statistical arbitrage strategies are also known as mean-reverting or relative value strategies. A critical point must be noted here, as statistical arbitrage is grounded in statistical regularities, assumptions on the existence of fair value amongst assets can be relaxed and in most cases are found to be irrelevant.

A *simple* statistical arbitrage strategy is pairs trading which aims to capitalize on the imbalances between two assets in the hope of making money once the imbalance is corrected. Pairs trading is a pure relative value strategy between two (possibly more) assets: Given two securities which historically moved together, take a long-short position as they diverge and realize a profit as the spread, the mispricing dynamics, converge back to the long-run mean.

Cointegration analysis is used to identify stochastic trends between securities. This is enhanced with an error-correction model (ECM) to provide a mechanism to capture short-term dynamics describing how the long-term equilibrium can be corrected to and restored. Though not proposing the theory, we demonstrate that if the markets were indeed efficient, predictable components could be identified given the right techniques.

Following from Kakoullis (2010), a random walk (RW) with a drift component is given by:

$$\gamma_t = a + \gamma_{t-1} + \varepsilon_t \quad (13.2)$$

where $\gamma_t \sim I(1)$, $\varepsilon_t \sim I(0)$ which can be seen as a first-order autoregressive model AR(1):

$$\gamma_t = a + \varphi\gamma_{t-1} + \varepsilon_t$$

where $\varepsilon \sim N(0, \sigma^2)$ with $\varphi = 1$ indicative of stationarity. We use the standard notation $I(1)$, integrated order 1, to denote that the process is non-stationary and needs to be differenced once in order to become stationary. Such a process is said to have a stochastic trend. To introduce